

NEOSTATS RISK

AI-Powered Credit Risk Intelligence Platform

Home Credit Default Risk · AI Engineer Assignment · NeoStats

Intelligence. Innovation. Impact.

ML Risk Scoring

SHAP Explainability

NL-to-SQL Chatbot

Docker Deploy

The Problem

Banks face growing pressure to make credit decisions that are fast, accurate, and explainable.



Too Slow

Manual reviews take days — customers expect instant decisions.



High Default Rates

Poor risk scoring leads to bad loans and financial loss.



No Explainability

Black-box models fail auditors and regulators.



Analysts Can't Self-Serve

Every data question goes through IT, creating bottlenecks.

Our Solution

One end-to-end AI platform — from raw data to explainable, auditable credit decisions.



EDA

Deep portfolio insights



Talk-to-Data

Ask in plain English



ML Model

Predict default risk



Explainable AI

SHAP-powered reasons



Decision Rules

Human-readable policy

Clean UI · Dockerized · Runs with a single command

**NEOSTATS RISK**

Credit Risk Intelligence Platform

[Dashboard](#)[EDA Analytics](#)[Scoring Engine](#)[Talk-To-Data](#)[Model Diagnostics](#)[Repository Explorer](#)

TOTAL APPLICATIONS

15,000Loaded from *syflife*

PORTFOLIO DELINQUENCY

7.83%

Average Default Ratio

AVG REQUESTED CREDIT

\$601,372

In USD (Kaggle Synthetic)

AVG GROSS INCOME

\$175,605

Household Underwriting

Live Deployment Running

AI Credit Underwriting & Decisioning

Welcome to the ****AI-Powered Credit Risk Intelligence Platform****. Engineered using the classic Home Credit Default Risk dataset, this dashboard delivers automated real-time underwriting scoring, robust local ****SHAP-style Explainable AI****, derived business rules, and a fully featured Natural Language SQL chatbot to talk directly with the loan database.

**Underwriting**

Evaluate risk trends

**Talk to Data**

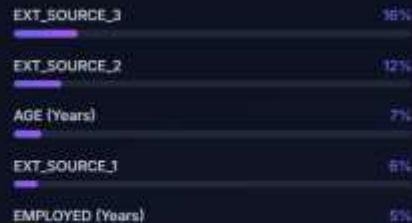
Query using NLP

**Model Metrics**

Review evaluation



Global Feature Importances





EDA Analytics — Portfolio Deep Dive

Live App Screen



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Model Diagnostics

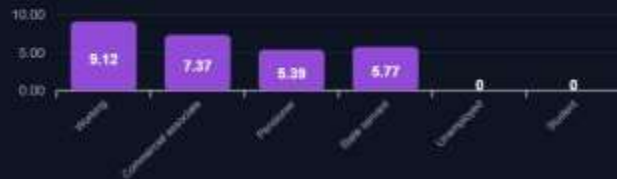
Repository Explorer

Portfolio Exploratory Data Analysis (EDA)

Default Rate (%) by Education Level



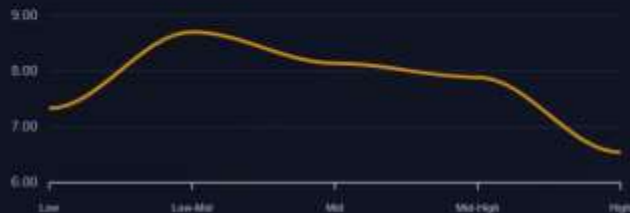
Default Rate (%) by Income Category



Top 8 Occupations by Delinquency Rate



Income Binning Margin Separation



Default rate by education, income, occupation · Income binning margin separation chart



Scoring Engine — Real-Time Risk Assessment

Live App Screen



NEOSTATS RISK

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Application Input

-- Load Preset Applicant --

Contract Type

Cash loans

Client Gender

Female

Owns Car

No

Owns Realty

Yes

Income Type

Working

Education Level

Secondary / Secondary Special

Annual Income (USD)

150000

Requested Credit (USD)

600000

Annual Annuity (USD)

30000

Goods Price (USD)

600000

Age (Years)

40

Employment (Years)

8

Ext Score 1

0.5

Ext Score 2

0.5

Ext Score 3

0.5

Run AI Scorecard Analysis



Awaiting Scorecard Trigger

Modify applicant demographic parameters on the left and trigger the AI scorecard. Real-time scoring, explainability vectors, and underwriting policies will display instantly.

Input applicant details → Run AI Scorecard Analysis → Get risk band (Low / Medium / High) instantly

Talk-to-Data — NL-to-SQL Chatbot

Live App Screen



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Secured SQL Agent Terminal

Translating Natural Language to SQL in Sandbox

Read-Only Sandbox



Hello, risk analyst! I am your conversational SQL database agent. You can ask me any question about our credit portfolio in plain English, and I will generate the correct SQL query, execute it, and provide analytical business insights!

TRY CLICKING ONE OF THE SUGGESTED PROMPTS ON THE RIGHT!

Ask a risk analysis question... (e.g. 'Show average income by default status')



Suggested Queries

Gender Distribution

Compare delinquency rates of Male vs Female applicants.

Income & Defaults

Calculate average income offsets for repeat vs default loans.

Occupational Risk Indices

Identify occupation fields carrying the highest risk.

Leverage Ratio Analysis

Analyze loan size leverage coefficients by income types.

Historical Application Pipeline

Break down approved vs denied count statistics.

5 pre-built queries: Gender distribution · Income & defaults · Occupational risk · Leverage ratio · Historical pipeline



Model Diagnostics & Performance Evaluation

Live App Screen



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Model Diagnostics & Performance Evaluation

ROC Curve (Sensitivity vs Specificity)



Precision-Recall (PR) Curve



Out-Of-Sample Confusion Matrix

True Negative

2454

False Positive

311

Underwriting Classifier Metadata

Classifier Architecture

Random Forest Classifier

Hyperparameters

estimators: 150, depth: 10, weights: balanced

Repository Explorer — Architecture Viewer

Live App Screen



NEOSTATS RISK

Credit Risk Intelligence Platform

 Dashboard

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 Repository Explorer

Repository Architecture Viewer

Interactive visual project blueprint

[VS Code Cyber Theme](#)

credit_risk_platform/

data/

(Mounted inside Docker container - not committed to git)

 Home Credit dataset files

documents/

 project_presentation.pdf

Project presentation (PDF)

notebooks/

 eda.ipynb

Exploratory Data Analysis

 eda.py

Converted ipynb to .py

src/

data/

 loader.py

Load and join dataset tables

 preprocessor.py

Cleaning, encoding, imputation

ml/

 train.py

Model training pipeline

 predict.py

Inference and scoring

 evaluate.py

Metrics: ROC-AUC, PR-AUC

talk_to_data/



File Inspector Active

Click on any folder to expand/collapse directories, or select a specific script or configuration file to view its full architectural details and dynamic workspace source code.

Interactive VS-Code style project blueprint · Click any file to view its architecture details and source

Machine Learning Layer — How It Works

Algorithm	Random Forest Classifier (150 estimators, depth 10)
Class Imbalance	class_weight: balanced — ensures minority class not ignored
Feature Engineering	120+ features: demographics, bureau data, credit history
Validation	Out-of-sample confusion matrix · ROC-AUC · PR-AUC
Output	Risk Score 0–100 → Low / Medium / High band

🔍 **LOW** · Score 0–35
Approve

🔍 **MEDIUM** · Score 36–65
Manual Review

● **HIGH** · Score 66–100
Decline

Major Design Decisions

01

Random Forest over Deep Learning

Interpretability and speed matter more than marginal accuracy gains. Random Forest produces reliable probability outputs and is directly compatible with SHAP — making it audit-ready and production-safe.

02

SQLite for Development, PostgreSQL-ready

SQLite keeps setup friction near zero — the evaluator doesn't need a database server. The data layer is abstracted so swapping to PostgreSQL in production requires only a config change.

03

LLM-Powered NL-to-SQL with Guard Rails

Natural language queries are converted via LLM prompts with strict READ-ONLY sandbox enforcement. Prompt templates are versioned to prevent hallucinations and SQL injection.

04

`class_weight: balanced` — No SMOTE Overhead

Oversampling (SMOTE) adds training complexity and can introduce noise. Balanced class weights handle the 8% default imbalance cleanly without synthetic samples, keeping the pipeline lean.

Architecture & Tech Stack

UI Layer · Multi-tab web dashboard (Dashboard · EDA · Scoring · Talk-to-Data · Diagnostics · Repo Explorer)

Application Layer · FastAPI backend · LLM NL-to-SQL agent · SHAP explainer · Decision rules engine

ML Layer · Random Forest Classifier · Feature engineering pipeline · Model training & inference

Data Layer · SQLite (dev) / PostgreSQL (prod) · Home Credit CSV · Preprocessing & imputation

ML

Random Forest · Scikit-learn

XAI

SHAP · LIME

LLM

Claude / OpenAI API

Backend

Python · FastAPI

Frontend

Streamlit / React

Deploy

Docker · Compose

```
$ git clone <repo> && docker-compose up → Full platform running in minutes
```

What We Built



15,000 applications analyzed · 7.83% default rate identified



Random Forest with 120+ features · balanced class weights



SHAP explainability on every prediction — audit-ready



NL-to-SQL chatbot · 5+ verified working queries



Dockerized · runs with a single command



Clean modular code · well-documented README

**Intelligence.
Innovation.
Impact.**

Thank You